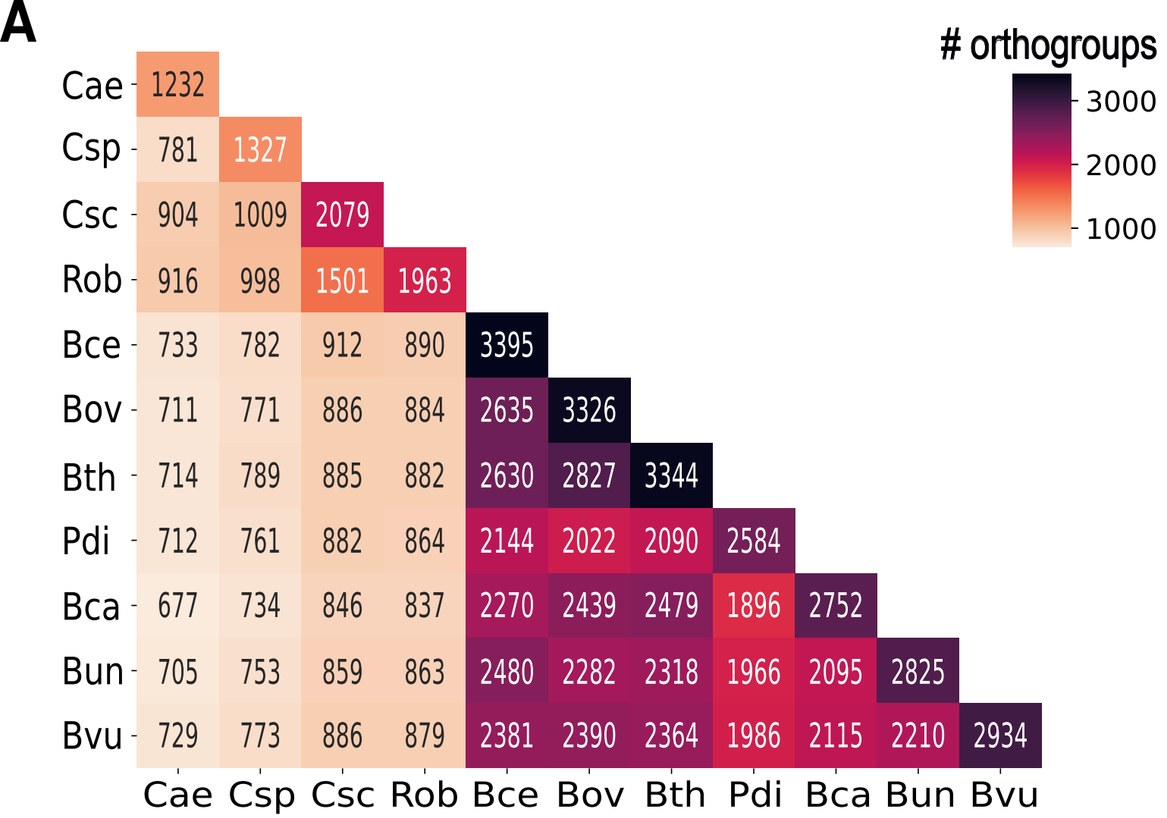
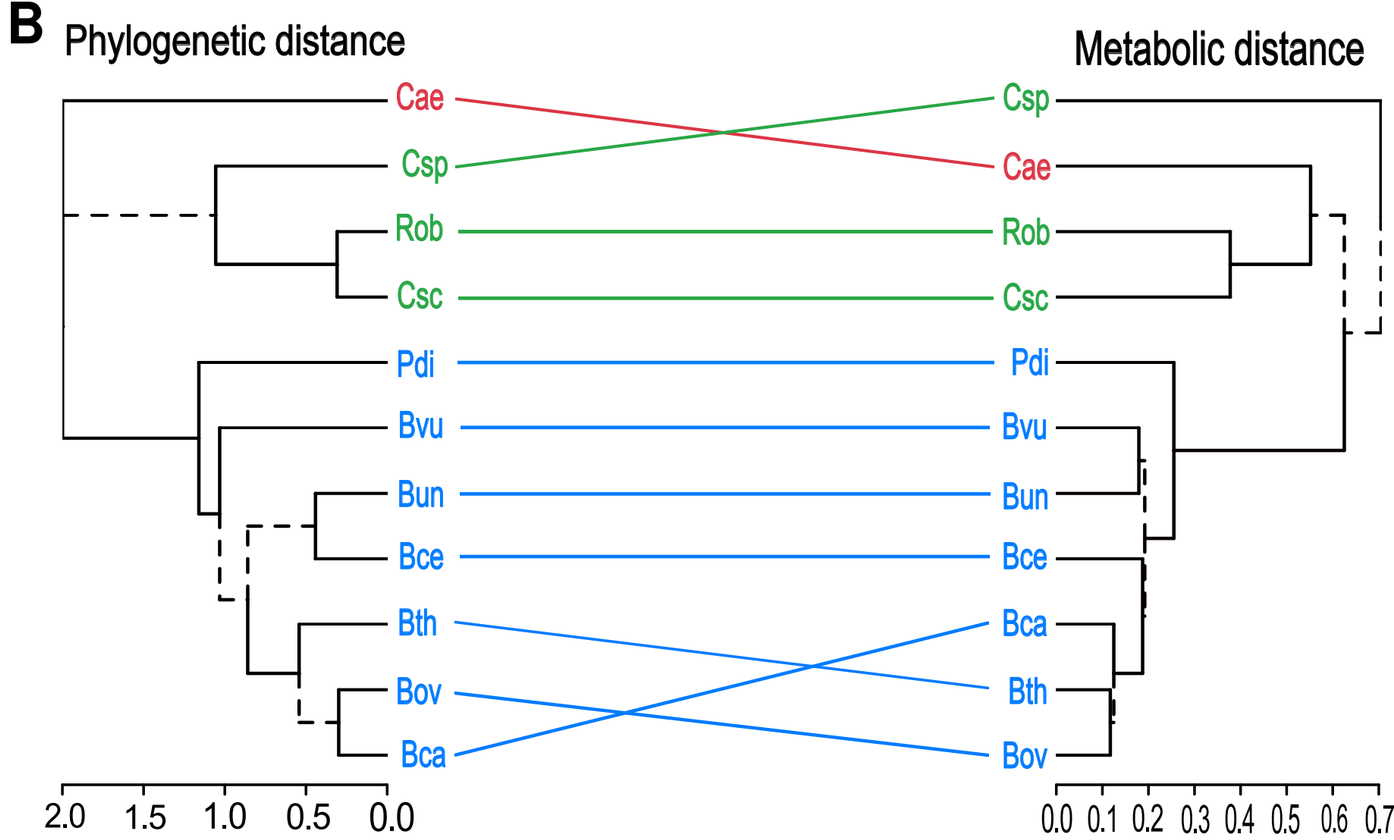
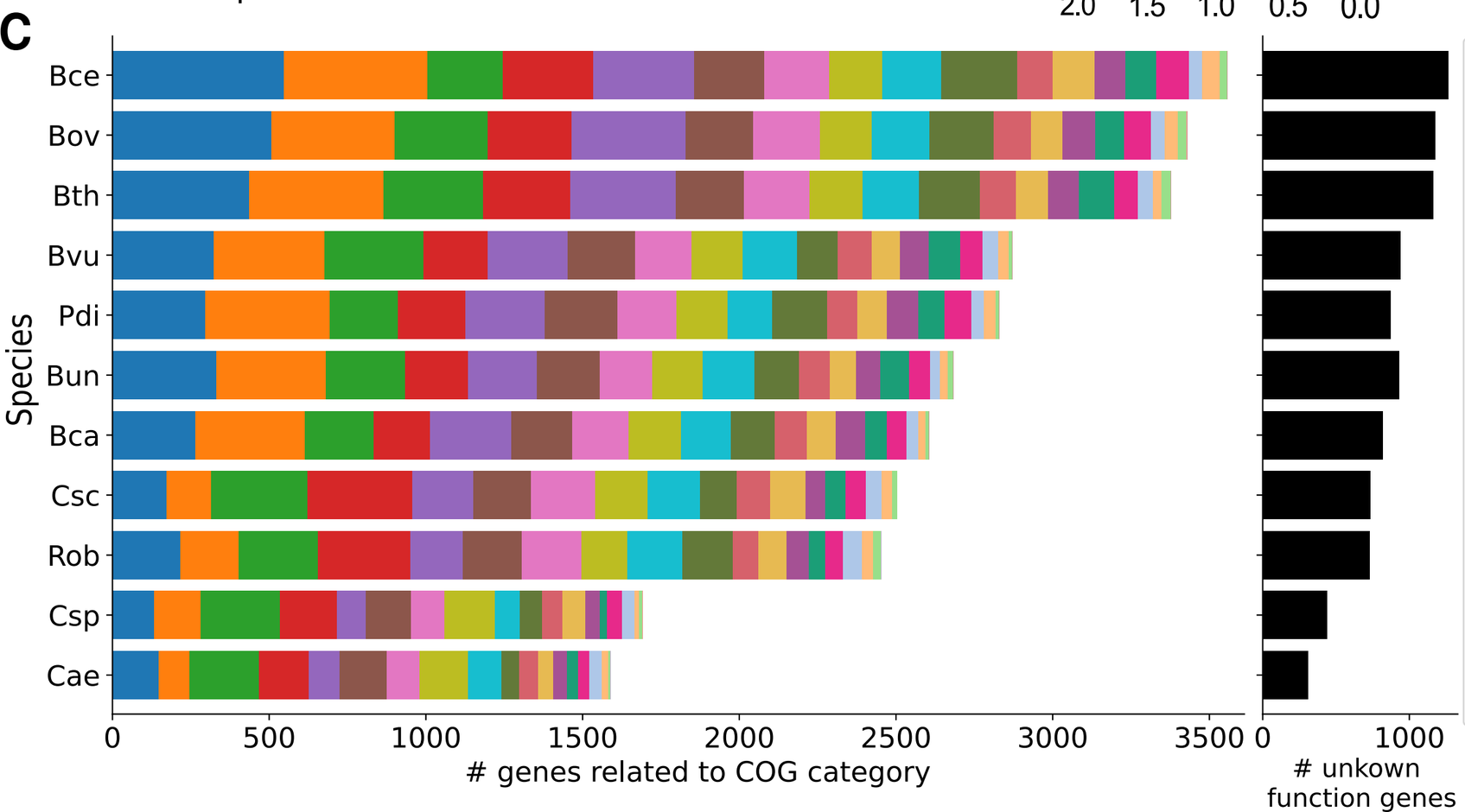
A

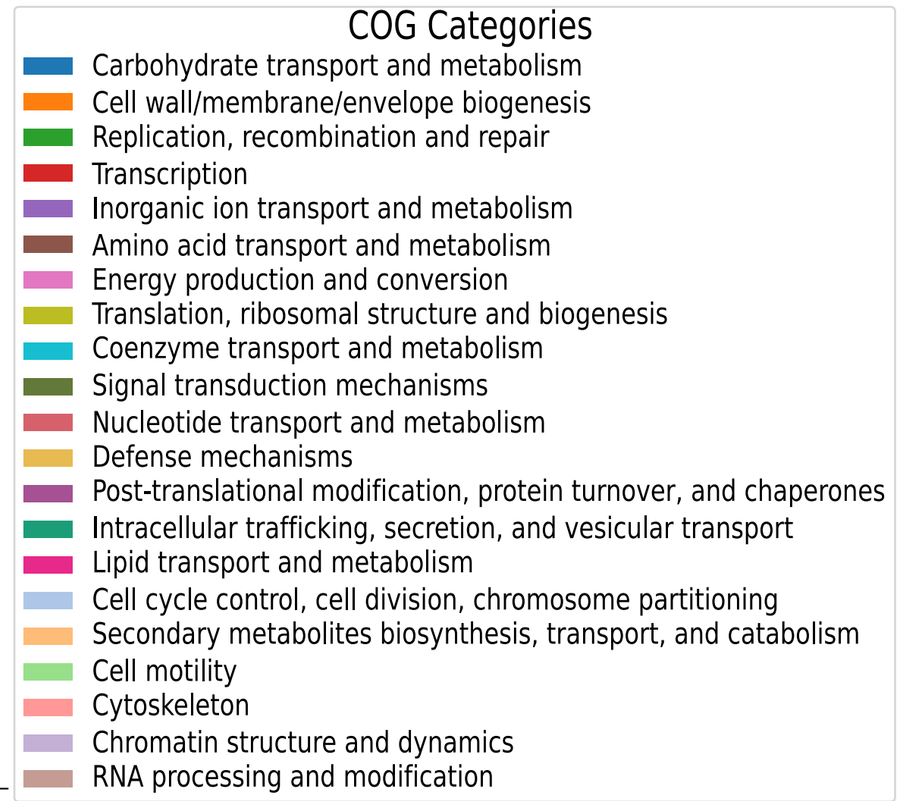

orthogroups
3000
2000
1000
Cae 1232
Csp 781 1327
Csc 904 1009 2079
Rob 916 998 1501 1963
Bce 733 782 912 890 3395
Bov 711 771 886 884 2635 3326
Bth 714 789 885 882 2630 2827 3344
Pdi 712 761 882 864 2144 2022 2090 2584
Bca 677 734 846 837 2270 2439 2479 1896 2752
Bun 705 753 859 863 2480 2282 2318 1966 2095 2825
Bvu 729 773 886 879 2381 2390 2364 1986 2115 2210 2934
Cae Csp Csc Rob Bce Bov Bth Pdi Bca Bun Bvu

B


Phylogenetic distance
Metabolic distance
Cae
Csp
Rob
Csc
Pdi
Bvu
Bun
Bce
Bth
Bov
Bca
2.0 1.5 1.0 0.5 0.0
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7

C


Species
Bce
Bov
Bth
Bvu
Pdi
Bun
Bca
Csc
Rob
Csp
Cae
0 500 1000 1500 2000 2500 3000 3500
genes related to COG category
0 1000
unkown function genes


COG Categories
Carbohydrate transport and metabolism
Cell wall/membrane/envelope biogenesis
Replication, recombination and repair
Transcription
Inorganic ion transport and metabolism
Amino acid transport and metabolism
Energy production and conversion
Translation, ribosomal structure and biogenesis
Coenzyme transport and metabolism
Signal transduction mechanisms
Nucleotide transport and metabolism
Defense mechanisms
Post-translational modification, protein turnover, and chaperones
Intracellular trafficking, secretion, and vesicular transport
Lipid transport and metabolism
Cell cycle control, cell division, chromosome partitioning
Secondary metabolites biosynthesis, transport, and catabolism
Cell motility
Cytoskeleton
Chromatin structure and dynamics
RNA processing and modification